

Do Gaussian Splats Keep the Nuclei?

An Object-Level Audit of Gaussian Volume Compression for Fluorescence Microscopy

[TODO: author names]

[TODO: affiliation]

Abstract—Gaussian splats are being adopted to store and stream large 3D microscopy and medical volumes, with budgets chosen and quality reported by PSNR. We ask whether, at a given number of bytes on disk, standard analysis still finds the nuclei. On 12 densely annotated frames from two *C. elegans* embryos (2,958 labelled nuclei), we pair each of 80 fits of a Gaussian-splat compressor (Luxar) with JPEG2000 encoded to exactly the same bytes, and score four detectors against human labels: a blob detector, a classical 3D watershed, and Cellpose in its 2D-stitched and 3D modes. Below $\approx 300\times$ the answer depends on the detector: the blob detector and 3D Cellpose find slightly more nuclei in splat reconstructions (+1.2 and +3.7 points on average), while the watershed and 2D Cellpose find more in JPEG2000 (−4.2 and −11.7) and never significantly fewer. Beyond that, at $\lesssim 3$ splats per nucleus, splats lose nuclei abruptly under every detector while JPEG2000 degrades gently. PSNR favours JPEG2000 in all 80 of 80 pairs and predicts none of this. Gaussian volume compression should be evaluated at matched bytes, against a verified codec, with the detector that will be used.

I. Introduction

Light-sheet and confocal time-lapse microscopy produce terabytes per experiment, and medical archives face the same pressure. Representing a volume as a set of anisotropic 3D Gaussians [1] is attractive for such data: it is compact, continuous and renders directly on the GPU. Recent systems fit Gaussian splats to microscopy volumes at scale [2], to slice-based microscopy and ultrasound [3], to medical volumes for compact storage [4], [5], and to tomographic reconstruction [6], [7]. Quality is measured by PSNR or SSIM throughout, and budgets are chosen by PSNR: Luxar selects its splat count by self-supervised held-out PSNR [8] and describes its fits as intended for visualisation rather than quantitative analysis [2].

What a microscopist needs from a compressed volume is that the analysis still works. Whether a nucleus remains detectable hinges on a small, often faint peak. Radiology learned this for lossy codecs, where acceptability is judged by whether lesions remain detectable, not by PSNR [9]. Gaussian volume compression has not been held to that standard, nor compared with standard codecs at identical stored bytes.

Contributions. (i) A rate–detectability protocol: identical bytes on disk, dense human labels, four detectors, paired bootstrap intervals over nuclei, and pass/fail rules committed before each new measurement, including a

replication on eight further frames. (ii) Whether splats or a standard codec keep more nuclei depends on the detector: volumetric detectors gain slightly from splats at moderate ratios, slice-wise and threshold-based segmenters lose—and the same network changes sides between its 2D and 3D modes. (iii) A budget cliff at $\lesssim 3$ splats per nucleus on every frame, invisible to PSNR. (iv) A baseline pitfall: a common JPEG2000 binding silently encodes a 3D array along the wrong axes, which had reversed our own high-ratio conclusion.

II. Related Work

Gaussian mixtures have long modelled nuclei for segmentation and lineage tracking [10]; our question concerns Gaussians as a storage format. Gaussian splatting [1] has been adapted to CT [7], [6], to slice-to-volume MRI with closed-form point-spread functions [11], and to volume compression [2], [3], [4], [5], each evaluated by image-fidelity metrics. JPEG2000 [12] and JPEG-XL [13] are the natural baselines at matched size. We use Cell Tracking Challenge labels [14], [15], a scale-normalised Laplacian of Gaussian [16], a distance-transform watershed, and Cellpose [17].

III. Methods

Data. Fluo-N3DH-CE [15]: *C. elegans* embryos imaged at $1.0 \times 0.09 \times 0.09 \mu\text{m}$ (8-bit, ≈ 12 MB per frame), every nucleus labelled. We use six frames per embryo (t110–t194 of sequence 01, t110–t185 of sequence 02; 102–362 nuclei each): four chosen first, eight added under rules fixed in advance. Embryo 1 t100, whose early nuclei are larger, is reported separately. Nuclei measure $3.6 \mu\text{m}$; all spatial parameters are fixed multiples of this diameter D .

Compressors. Luxar [2] with default settings, 1000 iterations and anisotropic voxel size, at seed budgets K from 500 to 6.4×10^4 ; its size is the .gsplats.zarr on disk, metadata included. Renders are mapped to raw intensities by a least-squares gain and offset (two numbers, not counted, which slightly favours Luxar). We also run Luxar’s own calibration—blind-spot held-out PSNR [8]—for its recommended K^* . JPEG2000 (OpenJPEG) codes each xy slice with a 2D wavelet, slices as components of one codestream, bisected to exactly the bytes of every

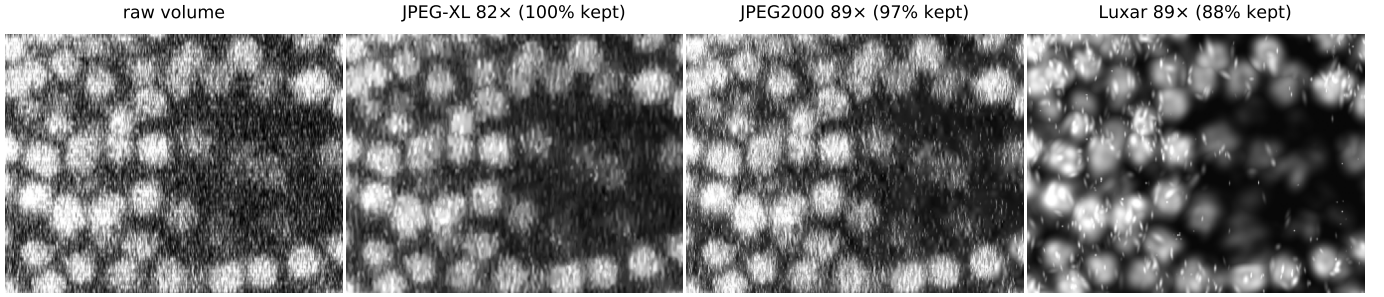


Fig. 1. The same region of embryo 1, t194 (3-slice maximum projection) from the raw volume, JPEG-XL, and JPEG2000 and Luxar at identical bytes; titles give whole-frame nuclei kept by 2D Cellpose. Both codecs preserve the textured appearance of nuclei; Luxar renders smooth blobs on a black background with thin streaks from elongated splats.

Luxar fit and to 50–800 KiB; JPEG-XL codes each slice, bisected to 50–800 KiB.

A baseline pitfall. Passing a (z, y, x) array to a common JPEG2000 binding encodes a z - y image with one component per x column, so the wavelet never runs along x . That configuration, which we used at first, costs ≈ 10 dB at $220\times$ and made splats appear far better than JPEG2000 beyond $200\times$; a regression test now checks the codestream header.

Detectors and scoring. (a) LoG: local maxima ranked by scale-normalised LoG response ($\sigma \in \{0.2, 0.27, 0.33, 0.4\}D$), top N kept, N = number of labels. (b) Watershed: background-subtracted, Otsu-thresholded foreground split by a distance-transform watershed in physical units, parameters fixed on raw volumes. (c) Cellpose 3 nuclei, 2D per slice stitched across z , and (d) the same model in its 3D mode (12 frames). On raw volumes they find 94%, 89%, 75% and 82% of the labelled nuclei. Detections match labels one-to-one within $0.6D$ (Table II varies it). Nuclei kept = nuclei found after compression / found in the raw volume by the same detector. Size-matched differences carry 95% bootstrap intervals over labelled nuclei, resampled jointly for raw and both reconstructions; they are per pair and not corrected for multiple comparisons, so we report counts of pairs. Pass/fail rules were committed before each new measurement [TODO: repository link]; the two that failed are reported in narrowed form.

IV. Results

Splats vs. JPEG2000 at identical bytes (Fig. 2, Table I). Below $300\times$, LoG keeps +1.2 points more nuclei in Luxar reconstructions on average (Luxar significantly better in 10 of 38 pairs, JPEG2000 in 4) and 3D Cellpose +3.7 (14 vs. 2 of 36). The watershed and 2D Cellpose go the other way: -4.2 and -11.7 points, with JPEG2000 significantly better in 22 of 38 and 23 of 38 pairs and Luxar never. Beyond $300\times$ (319 – $603\times$) every detector favours JPEG2000 (-16.8 , -19.8 , -23.5 and -18.7 points); JPEG2000 still keeps 98–103% (LoG) and 96–100% (watershed) of nuclei there. PSNR is higher for JPEG2000 in all 80 of 80

TABLE I
Luxar minus JPEG2000 at identical bytes: nuclei kept (points, mean over pairs; superscript: frames) and PSNR (dB). Bold: every pair’s 95% interval on the same side of zero.

Ratio	Spl./nuc.	Δ PSNR	LoG	Wshed	CP 2D	CP 3D
$\approx 30\times$	115–226	-6.5	$+1.7^3$	-2.2^3	-5.4^3	$+2.6^2$
$\approx 50\times$	57–115	-4.1	$+1.4^4$	-3.0^4	-10.5^4	$+6.1^4$
$\approx 90\times$	29–103	-2.7	$+1.8^{12}$	-2.7^{12}	-16.0^{12}	$+4.7^{12}$
$\approx 140\times$	16–30	-1.9	-0.2^4	-5.0^4	-11.8^4	$+4.5^4$
$\approx 230\times$	5–24	-1.3	$+0.7^{12}$	-6.2^{12}	-9.6^{12}	$+2.1^{12}$
$\approx 350\times$	2–10	-1.2	-3.4^{12}	-10.3^{12}	-12.6^{12}	-3.1^{12}
$\approx 480\times$	0.4–6.1	-1.6	-21.8^{12}	-22.9^{12}	-27.6^{12}	-24.7^{12}

pairs, by 1.0–6.8 dB. JPEG-XL is nearly lossless up to its maximum distance (82 – $120\times$; worst case 91%).

Why the detector matters. Smooth splats act as a denoiser for detectors that pool over a volume: under LoG and 3D Cellpose nuclei kept often exceeds 1 (Luxar at $45\times$ to $94\times$: 98–106% and 104–118%). Per-slice and threshold-based segmenters instead see smooth blobs on a black background with streaks (Fig. 1). The same Cellpose network shows this cleanly: against JPEG-XL at $45\times$ to $94\times$, splats cost it +4 to +100 points in 2D mode (intervals above zero on 4 of 4 frames) but gain it -18 to -4 in 3D mode, where the 5-point penalty we tested for in advance appeared on 0 of 12 frames—the penalty belongs to slice-wise processing, not to the learned model. The watershed sits in between (+0 to +5 points; under the pre-set 5 points on 11 of 12 frames).

The budget cliff and PSNR. Splats lose nuclei abruptly once they become scarce (Fig. 3): under LoG, nuclei kept crosses 90% at 1.1–3.1 splats per labelled nucleus on all 12 frames, as pre-registered (≤ 3.5). Over budgets $K \geq 1000$, Luxar’s PSNR varies by at most 2.6 dB per frame (19.0–23.2 dB overall). Our pre-registered test that PSNR misses the cliff within $K = 1000$ – $16\,000$ held on only 4 of 8 added frames: on smaller frames the cliff lies below that range. Survival—the fraction of the raw volume’s 100 strongest LoG detections still found, which needs no labels—tracks nuclei kept across fits (Spearman $\rho = 0.66$, $n = 80$) and can flag the cliff.

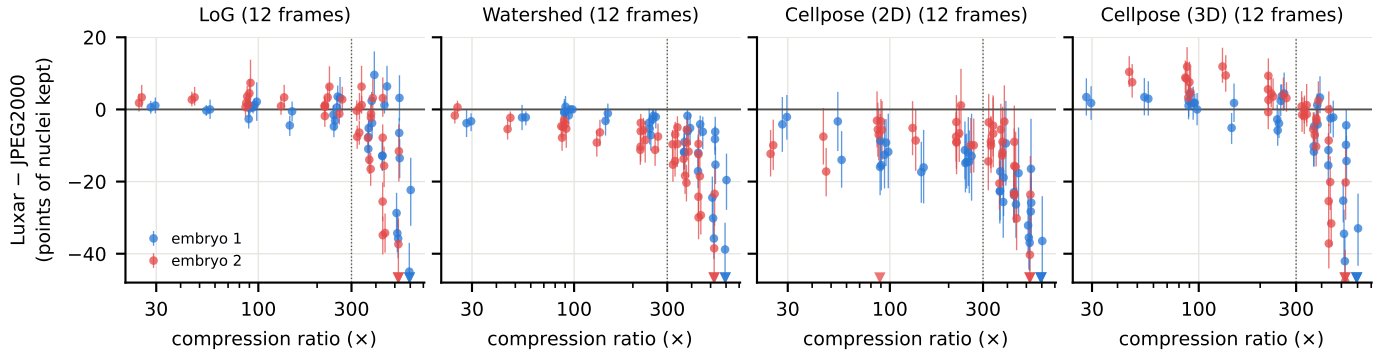


Fig. 2. Every Luxar fit against JPEG2000 at identical bytes: difference in nuclei kept (points; bars: 95% bootstrap intervals over nuclei) for four detectors. Above zero, splats keep more nuclei; triangles: below -48 points. Dotted line: $300\times$.

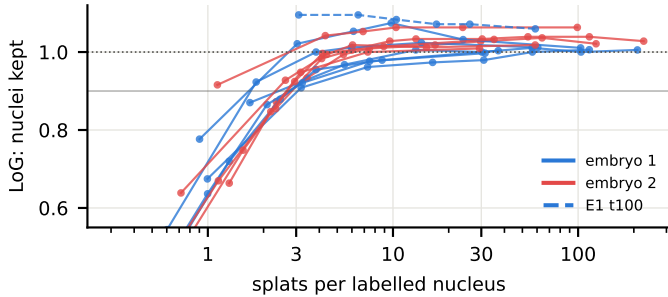


Fig. 3. The budget cliff: Log nuclei kept against splats per labelled nucleus, every Luxar fit on all frames (dashed: E1 t100).

TABLE II

Match-radius sensitivity (four first frames): nuclei found in the raw volumes and nuclei kept near $90\times$ (mean).

Radius	LoG			Cellpose (2D)		
	raw	JXL	Luxar	raw	JXL	Luxar
$0.4D$	1009	1.00	1.02	658	1.01	0.84
$0.6D$	1032	1.00	1.01	788	0.97	0.88
$0.8D$	1034	1.00	1.01	796	0.99	0.90

Luxar at its own budget. Luxar’s calibration recommends $K^* = 64,000$ on all 4 calibrated frames of both embryos—the top of our grid, held-out PSNR still rising. There, at $25\text{--}30\times$, it keeps 101–103% (LoG), 96–100% (watershed), 89–98% (2D) and 102–105% (3D Cellpose) of nuclei; JPEG2000 at the same bytes keeps 99–102%, in seconds rather than 17–46 min of fitting per volume on an A100 (5–40 min on an H100).

Reproducibility. Three independent Luxar fits of E1 t194 at each of three budgets differ by at most 2.1 (LoG), 3.5 (watershed) and 3.7 (2D Cellpose) points; Cellpose’s own scores move by up to 8 points between neighbouring codec sizes. On E1 t100, with larger early nuclei, the blob detector favours splats more strongly (8–11 points at $159\text{--}537\times$).

V. Discussion and Limitations

On this data no storage format is best for every analysis. JPEG-XL is nearly lossless to $\approx 100\times$; beyond it, JPEG2000 is the safer choice for segmentation and the only safe one past $300\times$, while Gaussian splats help volumetric detectors a little at moderate ratios, at the cost of minutes of GPU fitting. PSNR ranks the two formats the same way at every size and is therefore no guide to any of these outcomes. We recommend comparing Gaussian volume compression with a verified codec at identical bytes, reporting object-level recall with the detector that will be used downstream, and checking a splat budget at the object level before trusting it (survival is a label-free start).

Limitations: one organism (two embryos, 12 frames); one Gaussian compressor at default settings (tuned or entropy-coded splats may fare better); a 2D-per-slice JPEG2000 (a 3D wavelet would likely be stronger still); uncorrected per-pair intervals; and a fixed D that suits early, larger nuclei (t100) less well. The same audit for small low-contrast lesions in CT is future work.

VI. Conclusion

Measured at identical bytes against human labels, Gaussian-splat compression keeps nuclei well at its own budget, gains or loses a few percent against JPEG2000 at moderate ratios depending on the detector, and loses nuclei abruptly below a budget that PSNR cannot identify. Object-level, size-matched, detector-specific evaluation should accompany Gaussian volume compression.

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